



MASTG

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MASTG-TEST-0312: References to Explicit Security Provider in Cryptographic APIs

Overview

Android cryptography APIs based on the Java Cryptography Architecture (JCA) allow developers to specify a [security provider ↗](#) when calling `getInstance` methods. However, explicitly specifying a provider can cause security issues and break compatibility because several providers have been deprecated or removed in recent versions. For example:

- Apps targeting Android 9 (API level 28) or above [fail when a provider is specified ↗](#).
- The *Crypto* provider was deprecated in Android 7.0 (API level 24) and [removed in Android 9 \(API level 28\) ↗](#).
- The *BouncyCastle* provider was [deprecated in Android 9 \(API level 28\) and removed in Android 12 \(API level 31\) ↗](#).

This test identifies cases where an app explicitly specifies a security provider when using JCA APIs that is not the default provider, `AndroidOpenSSL` ([Conscrypt ↗](#)), which is actively maintained and should generally be used. It examines `getInstance` calls and flags any use of a named provider other than legitimate exceptions such as `KeyStore.getInstance("AndroidKeyStore")`.

Steps

- Run [Static Analysis on Android ↗](#) with a tool such as [semgrep ↗](#) on the app binary to look for calls to `getInstance` that explicitly specify a security provider.

Observation

The output should contain a list of locations where a security provider is explicitly specified in `getInstance` calls.

Evaluation

The test case fails if any `getInstance` call explicitly specifies a security provider other than `AndroidKeyStore` for `KeyStore` operations. Review each occurrence to determine whether the provider is actually required and whether its use could introduce security or compatibility issues on modern Android versions.

Best Practices

MASTG-BEST-0020: Update the GMS Security Provider

Demos

MASTG-DEMO-0075: Uses of Explicit Security Providers in Cryptographic APIs with semgrep